Ke Xueyi

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Education

Nanyang Technological University 08/2023 - 02/2025
M.Sc in Computer Control and Automation | GPA: 4.5/5.0 *Singapore*

Wuhan University 09/2019 - 06/2023
B.Eng in Electrical Engineering and Automation *Wuhan, China*

Publications

- **Xueyi Ke**, Satoshi Tsutsui, Yayun Zhang, and Bihan Wen. (2025). Discovering Hidden Visual Concepts Beyond Linguistic Input in Infant Learning. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. **[Accepted]**
- Winnie Pang, **Xueyi Ke**, Satoshi Tsutsui, and Bihan Wen. (2024). Integrating Clinical Knowledge into Concept Bottleneck Models. *Medical Image Computing and Computer Assisted Intervention (MICCAI)*. **[Accepted]**

Research Experience

Research Associate | Vision-Language-Action Models (VLA) Interpretability 08/2025 - Present
Nanyang Technological University *Advisor: Prof. Wen Bihan*

- Led mechanistic interpretability application on VLA models to decode the reasoning process from visual perception to action tokens.
- Identified and analyzed specific attention circuits and neurons to uncover mechanisms of learning dynamics in embodied AI.
- Investigating the recovery of induction heads to enable and enhance one-shot imitation learning capabilities in embodied agents.

Research Intern | Explainable Computer Vision in Multimodality 01/2024 - 12/2024
Nanyang Technological University *Advisor: Prof. Wen Bihan*

- Interpreting CLIP-like multi-modal models by decomposing each neuron's function, explaining neurons using 'dissection' techniques.
- Enhanced model explainability by integrating medical domain knowledge into concept bottleneck models (CBM).
- Achieved a 12%+ improvement in accuracy by applying data augmentation techniques like GANs and rectifying imbalance issues.

Research Intern | Multimodal Perception for Autonomous Driving 05/2023 - 08/2023
Tsinghua University *Advisor: Prof. Wang Li*

- Collaborated on the publication 'Object Perception for Autonomous Driving', focusing on single/multi-modal perception.
- Implemented algorithm on a TX2-based mini-unmanned vehicle, achieving over 88% accuracy in campus scene object detection.
- Contributed to constructing the 'Dual Radar' dataset tailored to monitor extreme weather conditions through data analysis techniques.

Industry Experience

Algorithm Research Intern 04/2025 - 08/2025
Alibaba Group Beijing, China

- Led project using multimodal congested chat data for spatio-temporal prediction, achieving 18% gain over SOTA.
- Built attention-based MM-ST fusion framework; applied SFT and LoRA to mine signals from sparse chat using LLMs and MLLMs
- Deployed multimodal features to ETA prediction in real-world setting scenario, achieving a 12.4% reduction performance in MAE.

AI Scientist Intern 05/2024 - 08/2024
TE Connectivity Ltd. AI Hub *Singapore*

- Led the project on AI-driven PCB design, developing a DRL algorithm to optimize component placement under EMI constraints.
- Implemented detection and segmentation algorithm for electric wire images, improving metrics by 20% by addressing class imbalance.

Projects

PassGPT - Personalized Education Agent 03/2024 - 06/2024

- Led pipeline design, defined scope and scenarios for educational QA; applied SFT and RLHF on ~100K domain-specific samples.
- Built an RAG agent with MySQL, AWS S3, and FAISS integration, deployed with Claude, GPT-4, and LLaMA models.
- Led the collection and construction of ~50K educational cleaned documents to support accurate and grounded chatbot responses.

Skills

- **Core Competencies:** Mechanistic Interpretability, VLA/LLM Training/Fine-tuning (vLLM, LoRA), Multi-modal Learning, RAG
- **Programming Skills:** Python (PyTorch), C/C++, Git, Bash, Zsh, Docker, SQL
- **Languages:** English (Advanced), Chinese (Native)